

Paolo Fiol

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AI and full-stack software engineer with 3+ years of experience turning complex operational workflows into production SaaS, LLM integrations, and agentic automation. Built EHR browser agents with TypeScript, Playwright, AWS, and Amazon Bedrock; shipped enterprise software with Angular and Node.js; and deployed a Next.js/PostgreSQL generative AI product. Strong in stakeholder discovery, rapid prototyping, REST API integration, structured outputs, evaluation, guardrails, cloud deployment, and production troubleshooting.

Work Experience

Independent Software Engineer

March 2026 - Present

AI Engineer, Beacon Health (Contract)

Jun 2026 - Sep 2026

- Translated a clinical pre-charting workflow into an AI-assisted EHR automation agent using TypeScript, Playwright, Node.js, AWS EC2/Systems Manager, and Amazon Bedrock; extracted chart data, populated visit-note templates, and drafted patient letters.
- Improved reliability by separating deterministic browser actions from schema-validated LLM output, diagnosing a lazy-loaded data integrity failure, and adding field-level provenance logging for auditable PHI workflows.
- Evaluated three agent and browser automation frameworks and documented implementation tradeoffs, failure modes, and rollout safeguards including sandbox validation, draft-only writes, identity verification, and constrained model outputs.

Software Engineer, Resume Reframe (Personal SaaS Project) - <https://resume-reframe.com/>

Mar 2026 - Present

- Built and deployed an AI-powered resume tailoring SaaS using TypeScript, Next.js 15, PostgreSQL, and Drizzle ORM; integrated Google Gemini with structured prompting to generate job-specific resumes and cover letters.
- Owned the product end to end, including PDF/DOCX parsing, email/password and Google OAuth authentication, application tracking, DOCX export, Redis rate limiting, Stripe billing, and Vercel deployment.

Software Engineer, Tata Consultancy Services

July 2022 - Oct 2025

- Led full-stack feature delivery for an enterprise workforce management platform, translating business requirements into AngularJS and Node.js releases while collaborating with product, QA, and stakeholder teams.
- Contributed to an enterprise AI assistant integrated into the workforce platform, supporting LLM-powered employee self-service workflows through integration testing, response validation, and production troubleshooting in a 60K+ user environment.
- Designed and optimized SQL queries, replacing legacy subqueries with joins to improve database performance and maintainability.
- Maintained automated CI/CD scripts and delivered an optimization that reduced test pipeline time by 50%, accelerating production delivery.

Skills

Programming Languages: TypeScript, JavaScript, Python, C#, Java, C++, SQL, HTML, CSS

Frameworks / Libraries: React, Next.js, Angular, Node.js, Express, Playwright, .NET, ASP.NET Core, Spring Boot, Drizzle ORM

AI / Automation: LLM integration, generative AI, AI agents, agentic workflows, prompt engineering, structured outputs, model evaluation, response validation, guardrails, browser automation, Amazon Bedrock, Gemini, OpenAI API, Claude

Cloud / Data / Delivery: AWS (EC2, IAM, Systems Manager, Bedrock), PostgreSQL, Redis, REST APIs, OAuth, Docker, Linux, Git, CI/CD, Azure DevOps, Vercel, Supabase, Stripe

Education

University of Florida - Bachelor of Science in Computer Science, GPA: 3.37

Graduated May 2022

- Coursework: Data Structures & Algorithms, Database Systems, Intro to Artificial Intelligence, Software Engineering